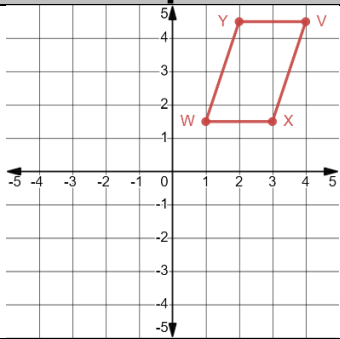
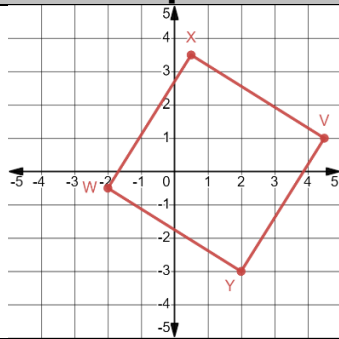


Geometry
Unit 13
Lesson 11 Homework

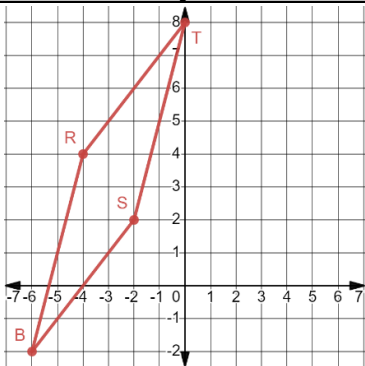
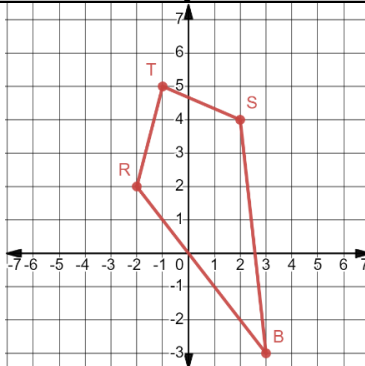
Name Answer Key
Date _____
Period _____

Plot the given coordinates in to order to compare each pair of graphs. Record your work and show your justifications in the table.

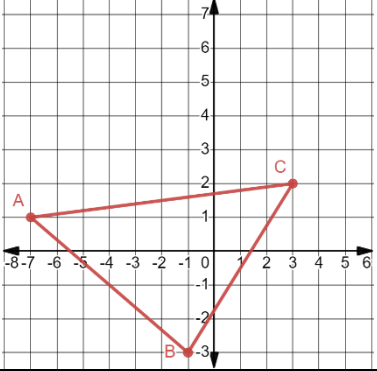
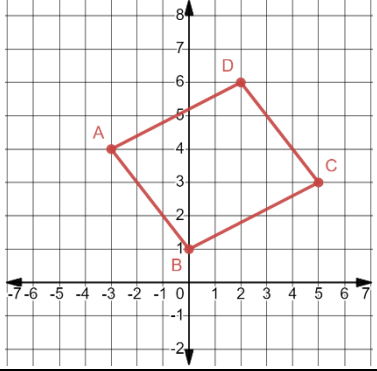
1.

Graph 1	Graph 2
	
Coordinates: $V(4,4.5), X(3,1.5), W(1,1.5), Y(2,4.5)$	Coordinates: $V(4.5,1), X(0.5,3.5), W(-2, -0.5), Y(2, -3)$
Classification: Parallelogram	Classification: Square
Property: Opposite sides are parallel and congruent. Diagonals bisect each other.	Property: All sides congruent. Opposite sides parallel. Diagonals congruent & bisect each other.
Justification: Slopes of $\overline{YV}$ & $\overline{WX}$ are zero. Slopes of $\overline{WY}$ & $\overline{XV}$ are 3. $WX = YV = 2$ $WY = XV = \sqrt{3^2 + 1^2} = \sqrt{10}$	Justification: $VX = XW = WY = YV = \sqrt{\left(\frac{5}{2}\right)^2 + 4^2} = \frac{\sqrt{89}}{2}$ Slopes $WX = YV = \frac{8}{5} \perp$ to slopes $WY = XV = -\frac{5}{8}$

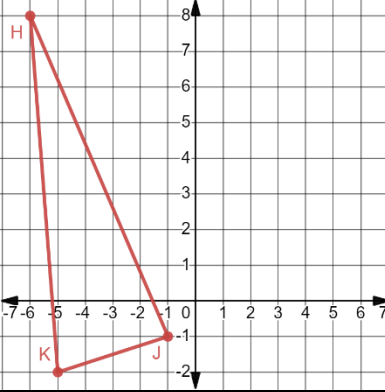
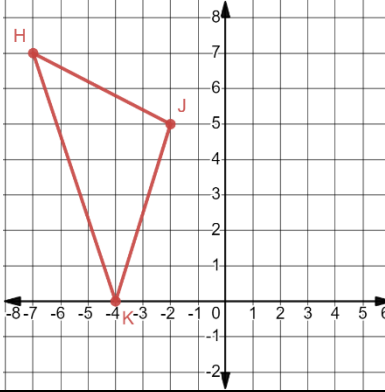
2.

Graph 1	Graph 2
	
Coordinates: $T(0,8), R(-4,4), B(-6, -2), S(-2,2)$	Coordinates: $T(-1,5), R(-2,2), B(3, -3), S(2,4)$
Classification: Parallelogram	Classification: Kite
Property: Opposite sides are parallel and congruent. Diagonals bisect each other.	Property: Perpendicular diagonals. Two pairs of congruent adjacent sides.
Justification: $BR = ST = 2\sqrt{10}, TR = SB = 4\sqrt{2}$ Slopes $BR = ST = 3$ Slopes $TR = SB = 1$	Justification: Slope $TB = -2$; Slope $RS = \frac{1}{2}$ $RB = SB = 5\sqrt{2}$ $RT = TS = \sqrt{10}$

3.

Graph 1	Graph 2
	
Coordinates: $A(-7, 1)$, $B(-1, -3)$, $C(3, 2)$	Coordinates: $A(-3, 4)$, $B(0, 1)$, $C(5, 3)$, $D(2, 6)$
Classification: Scalene triangle	Classification: Parallelogram
Property: No sides congruent.	Property: Opposite sides are parallel and congruent.
Justification: $AB = 2\sqrt{13}$ $BC = \sqrt{41}$ $AC = \sqrt{101}$	Justification: $Slopes\ AD = BC = \frac{2}{5}$ $Slopes\ AB = DC = -1$ $AB = DC = 3\sqrt{2}$; $AD = BC = \sqrt{29}$

4.

Graph 1	Graph 2
	
Coordinates: $H(-6, 8)$, $J(-1, -1)$, $K(-5, -2)$	Coordinates: $H(-7, 7)$, $J(-2, 5)$, $K(-4, 0)$
Classification: Scalene triangle	Classification: Isosceles right triangle
Property: No sides congruent.	Property: Exactly two sides are congruent. Two sides are perpendicular.
Justification: $HJ = \sqrt{5^2 + 9^2} = \sqrt{106}$ $JK = \sqrt{1^2 + 4^2} = \sqrt{17}$ $HK = \sqrt{1^2 + 10^2} = \sqrt{101}$	Justification: $HJ = \sqrt{2^2 + 5^2} = \sqrt{29}$ $JK = \sqrt{2^2 + 5^2} = \sqrt{29}$ $HK = \sqrt{3^2 + 7^2} = \sqrt{58}$ $Slope\ HJ = -\frac{2}{5}$; $Slope\ KJ = \frac{5}{2}$